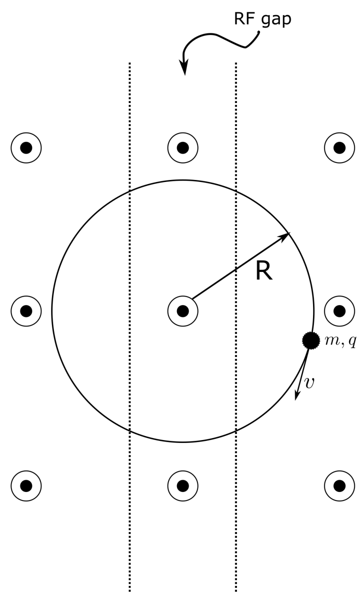
Instructions:

- Solutions to the problem sets are due by 9:00 a.m. the next morning.
- While points may differ between Problem Sets each Problem Set will carry equal weight for course grade.
- Please list your collaborators on your submission.

Problem 1. Cyclotron - Single Particle 20 pts.



a) 5 pts: Neglecting the change in orbit radius, R, over one transit, follow the steps in class to show that the non-relativistic period τ of the orbit is:

$$\tau = \frac{2\pi}{\omega_c}$$

where $\omega_c = \frac{qB}{m}$ is the cyclotron frequency and is constant.

Discuss the implications for a particle entering the oscillating RF field gap each transit. Will the frequency of the RF field need to change?

b) 5 pts: If a particle with mass, m, is traveling relativistically with approximately constant velocity, v, over one turn in a cyclotron, show that:

$$\frac{d\vec{p}}{dt} = -\frac{m\gamma v^2}{R}\hat{r}$$

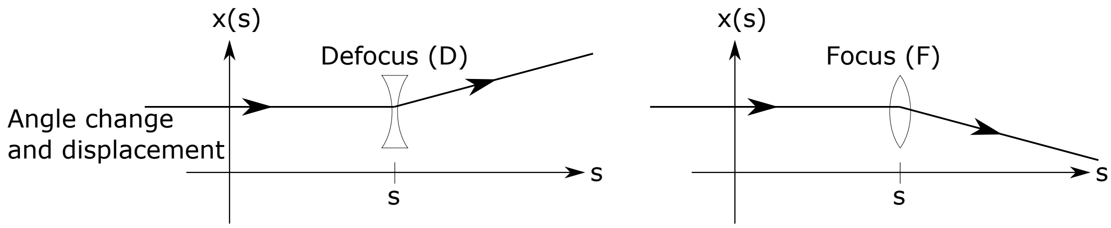
where $\vec{p} = \gamma m\vec{v}$ and $\gamma = 1/\left(1 - \frac{\vec{v}^2}{c^2}\right)^{\frac{1}{2}} \approx$ const.

c) 10 pts: Derive a relativistically correct formula for the period, τ, to characterize how it varies with particle kinetic energy, $\mathcal{E} = (\gamma - 1)\, mc^2$.

Discuss the implications of this result for the operation of a cyclotron as the particle becomes relativistic.

Problem 2. Thin Lens FODO Phase Advance and Stability 15 points

A thin lens changes the angle of a particle trajectory but not the coordinate:



This action can be specified by transfer matrices applied at $s = s_i$

$$\begin{bmatrix} x \\ x' \end{bmatrix} = \mathbf{M}(s|s_i) \begin{bmatrix} x_i \\ x_i' \end{bmatrix}$$

Focusing:

$$\mathbf{M}_F = \begin{bmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{bmatrix} \qquad f > 0$$

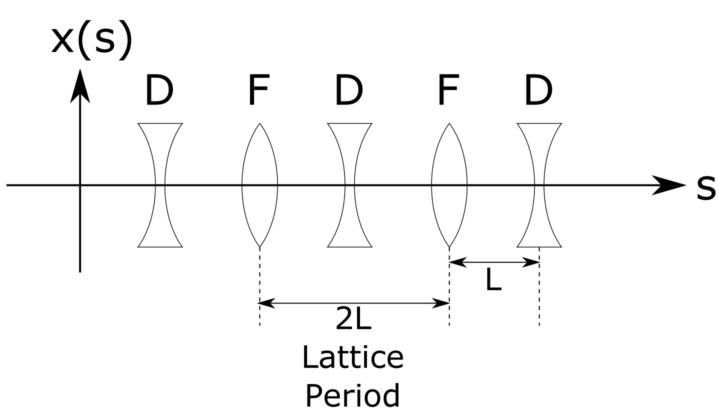
Defocusing:

$$\mathbf{M}_D = \begin{bmatrix} 1 & 0 \\ \frac{1}{f} & 1 \end{bmatrix} \qquad f > 0$$

A free space drift of length L has a transport matrix:

$$\mathbf{M}_0 = \begin{bmatrix} 1 & L \\ 0 & 1 \end{bmatrix}$$

Consider a lattice of period $2L$ made up of equally spaced F and D lenses with equal values of f.

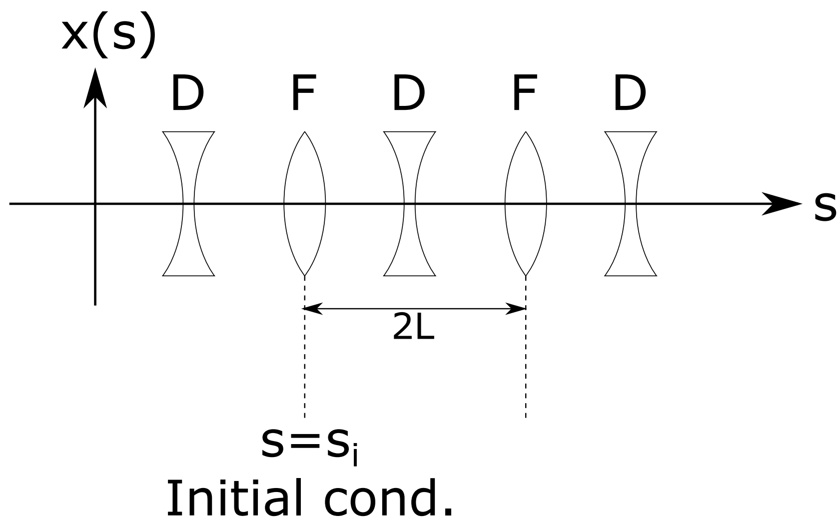


a) 10 pts: Calculate $\cos\Phi$ where Φ is the particle phase advance. Express the answer in terms of L/f. Also, determine the range of f for which the particle orbit is stable.

b) 5 pts: For the case of f chosen to correspond to the stability limit, sketch the motion of a particle with initial conditions:

$$\lim_{s \to s_i} x(s) = x_0$$
$$\lim_{s \to s_i} x'(s) = x_o/L$$

where $s = s_i$ is the axial location of a focusing thin lens kick, and $s \to s_i^-$ is just before the kick. Sketch the particle orbit for focusing strength slightly off from the stability limit into the unstable region. Superimpose the orbit sketch on a diagram of the lattice (see below):



Problem 3. Drift Twiss Parameters 20 pts.

Consider a long, field-free region adjacent to a collision point. The lattice is designed such that the beam size is minimized at the collision point, represented by $\beta = \beta^*$. Recall that the Twiss functions are propagated from s to s_0 using the transfer matrix:

$$\begin{bmatrix} \beta \\ \alpha \\ \gamma \end{bmatrix} = \begin{bmatrix} \mathrm{m}_{11}^2 & -2m_{11}m_{12} & m_{12}^2 \\ -m_{21}m_{11} & m_{11}m_{22} + m_{12}m_{21} & -m_{12}m_{22} \\ m_{21}^2 & -2m_{22}m_{21} & m_{22}^2 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \alpha_0 \\ \gamma_0 \end{bmatrix}$$

a) 5 pts: Use the transfer matrix propagation to show that the β-function evolves with s like

$$\beta(s) = \beta^* + \frac{(s - s_0)^2}{\beta^*}$$

b) 5 pts: How does $\alpha(s)$ evolve?

c) 5 pts: How does the phase $\psi(s)$ evolve?

d) 5 pts: What is the largest phase advance possible across a field-free region?